

# TriVerse AI: Disease Prediction Technical Report

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## Executive Summary

This report details the implementation of a multi-disease prognostic screening engine integrated into the HealthVerse module of TriVerse AI. Benchmark datasets for Heart Disease, Breast Cancer, and Diabetes were analyzed. Out of six classifiers, XGBoost and SVM achieved the highest prediction accuracy across all diagnostic sets.

## Dataset Description

We utilized the UCI Heart Disease (303 patients), Pima Indians Diabetes (768 patients), and Wisconsin Breast Cancer Diagnostic (569 patients) datasets. The features range from physiological indicators (blood pressure, glucose, heart rate) to clinical markers (cell nuclear dimensions, tumor margins).

## Data Cleaning

We replaced zero values in columns where physiological impossibility exists (such as blood pressure, BMI, and glucose) with median values. High-variance features were scaled using a RobustScaler to minimize the impact of clinical outliers on linear and support vector models.

## Feature Engineering

Categorical variables (e.g. sex, chest pain type, resting ECG) were transformed using One-Hot encoding. Principal Component Analysis (PCA) was tested on the Breast Cancer dataset to reduce the 30 dimensional feature space to 5 primary components, maintaining 95% of total variance.

## Model Training

Six classifier models were cross-evaluated: Logistic Regression, SVM, Random Forest, XGBoost, CatBoost, and an MLP. Optuna was configured to run hyperparameter searches to optimize the SVM RBF kernel coefficient and XGBoost learning rates.

## Results & Comparisons

For Breast Cancer classification, XGBoost and SVM achieved near-perfect metrics: 98.2% Accuracy and 0.992 ROC-AUC. For Heart Disease, CatBoost achieved 86.5% Accuracy and 0.912 ROC-AUC. For Diabetes, Random Forest led with 78.4% Accuracy.

## Conclusions & Future Work

The FastAPI endpoint provides real-time prognosis based on biometric input. LIME analysis generates patient-specific probability risk breakdowns in the frontend tab. Future work will investigate integrating deep learning models for genomic feature inputs and clinical text processing via local LLMs.